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Schema-based classification of data on a system

Abstract

A virtualized computing system includes a plurality of hosts, each of which is configured with a virtualization software for supporting execution of virtual machines therein. A method of managing a configuration of a system service in the virtualized computing system includes upon receiving an API call to operate on a configuration object for the system service that is backed by a configuration schema of the system service, updating a database in accordance with the configuration schema to update or store the configuration of the system service, so that the system service executes with the configuration stored in the database.

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Background/Summary

BACKGROUND

(1) A typical software stack for configuration management of a system includes an API (Application Programming Interface) layer, which provides an endpoint to configure and monitor the system, a business logic layer, which contains the API implementation, and a persistence layer, which persists any configuration or state change in the system onto a disk. In this system, it is not possible to figure out all of the configuration tasks performed by the end user while the system is live, especially after a long period of time has passed since boot-up of the system, because the configuration actions are not being persisted. Rather, the resulting state of those actions is persisted. As such, the system can only report the current state of the configuration, and it would not be possible to revert back to a certain configuration state. In fact, it is difficult to even revert back to the initial default configuration state.

(2) The inability to figure out the configuration tasks performed by the user is even a bigger issue when the system must be managed at scale, because the complexity of managing the system grows with number of configurations that must be set and monitored. As a result, only ad hoc solutions,

which provide configuration and compliance support for a limited set of configurations, are available in large-scale systems.

SUMMARY

(3) One or more embodiments provide a system in which every single property on the system which needs to be persisted is defined upfront in a configuration schema. The configuration schema classifies each such property as either configuration or state. A “configuration” is data that is provided by the user as part of a configuration action. A “state” is data that is internally generated in the system. The state is further classified into two types, vital or cached. A “vital” state is state data that must be persisted across reboots. A “cached” state is state data that does not need to be persisted across reboots.

(4) In one embodiment, the system is a virtualized computing system comprising a plurality of hosts, each of which is configured with a virtualization software for supporting execution of virtual machines therein. A method of managing a configuration of a system service in such a system includes: upon receiving an API call to operate on a configuration object for the system service that is backed by a configuration schema of the system service, updating a database in accordance with the configuration schema to update or store the configuration of the system service, so that the system service executes with the configuration stored in the database.

(5) Further embodiments include a non-transitory computer-readable storage medium comprising instructions that cause a computer system to carry out the above method, as well as a computer system configured to carry out the above method.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a simplified block diagram of a virtualized computing system of the prior art in which configurations are persisted in files.

(2) FIG. 2 is a simplified block diagram of a virtualized computing system according to embodiments in which configurations are persisted in key-value store in accordance with a configuration schema.

(3) FIG. 3A is an example of a schema definition.

(4) FIG. 3B is a configuration schema generated from the schema definition of FIG. 3A.

(5) FIG. 4A is another example of a schema definition.

(6) FIG. 4B is a configuration schema generated from the schema definition of FIG. 4A.

(7) FIG. 4C is a default configuration schema generated from the schema definition of FIG. 4A.

(8) FIG. 4D is an edited version of the default configuration schema of FIG. 4C.

(9) FIG. 5A illustrates the use of an API to create or overwrite a configuration.

(10) FIG. 5B illustrates the use of an API to update a configuration.

(11) FIG. 5C illustrates the use of an API to retrieve a configuration.

(12) FIG. 5D illustrates the use of an API to delete a configuration.

(13) FIG. 5E illustrates the use of an API to retrieve a default configuration.

(14) FIG. 5F illustrates the use of an API to overwrite a default configuration.

(15) FIG. 6 is a flow diagram that illustrates a system service that executes with a configuration that is set according to embodiments.

(16) FIG. 7 is a flow diagram that illustrates a method of resetting all system services to execute with a default configuration.

(17) FIG. 8 depicts entries in a key-value store after resetting of FIG. 7 is carried out.

DETAILED DESCRIPTION

(18) FIG. 1 is a simplified block diagram of a virtualized computing system of the prior art in which configurations are persisted in files that are stored in data store 30. The virtualized

computing system of the prior art includes a virtual machine (VM) management server **10** that manages the lifecycle of VMs running in a cluster of hosts **20**. To configure system services running in hosts **20**, an end user operates a user interface (UI) on VM management server **10** to make configuration API calls to hosts **20**, in particular host daemon **21** running in each host **20**. Examples of system services that are configured according to embodiments include network time service running on an NTP (network time protocol) server, a secure shell (SSH) service, an authentication service, a firewall service, network service, storage service, keyboard service, etc. Host daemon **21** processes the configuration API calls. If the API call is to set a configuration for a system service, host daemon **21** instructs configuration persistence layer **22** to persist the configuration in one of files, F1, F2, . . . , Fn, corresponding to the system service. If the configuration for the system service is persisted in file F1, for example, the next time the system service is executed, it is executed with the configuration stored in file F1.

(19) FIG. 2 is a simplified block diagram of a virtualized computing system according to embodiments in which configurations are persisted in a key-value store **140** provisioned in data store **130**. The virtualized computing system according to embodiments includes a VM management server **100** that manages the lifecycle of VMs running a cluster of hosts **120**. To configure system services running in hosts **120**, an end user operates a UI on VM management server **100** to make configuration API calls to hosts **120**, in particular host daemon **121** running in each host **120**. Configuration API calls includes a set API call (to set a configuration for a system service), an update API call (to update a configuration for a system service), a get API call (to retrieve a configuration for a system service), a delete API call (to delete a configuration for a system service), and a “get default configuration” API call (to retrieve a default configuration for a system service). Host daemon **121** receives the configuration API call issued by VM management server and passes it onto config store layer **122** for processing. Config store layer **122** exposes configurations for different system services as configuration objects, each backed by a corresponding configuration schema, and records all updates to the configurations of system services in a key-value store **140**. In key-value store **140**, the “key” corresponds to the system service and the “value” corresponds to one or more configuration properties of that system service. (20) If the configuration API call is the set API call, config store layer **122** creates or overwrites a configuration object corresponding to the system service identified in the set API call. FIG. 5A illustrates the use of the set API call to create a configuration object corresponding to the system service “ntp” and update key-value store **140** to add this configuration object as a new entry in key-value store **140**.

(21) If the configuration API call is the update API call, config store layer **122** updates a configuration object corresponding to the system service identified in the update API call. FIG. 5B illustrates the use of the update API call to update a configuration object corresponding to the system service “ntp” and update key-value store **140** to update this configuration object.

(22) If the configuration API call is the get API call, config store layer **122** retrieves a configuration object for the system service identified in the get API call from key-value store **140**. FIG. 5C illustrates the use of the get API call to retrieve a configuration object for the system service “ssh” and the retrieved result.

(23) If the configuration API call is the delete API call, config store layer **122** deletes a configuration object for the system service identified in the delete API call from key-value store **140**. FIG. 5D illustrates the use of the delete API call to delete a configuration object for the system service “ntp” from key-value store **140**. After this delete API call is processed, the entry for the key “ntp” remains in key-value store **140** because the internal state “drift” is defined in the configuration schema as a “vital” state.

(24) FIG. 3A is an example of a schema definition. The schema definition is written in a high-level language that permits the following annotations: @vital—Classifies data to be internal state data that must be persisted across reboots. @cached—Classifies data to be internal state data that is not

persisted across reboots. @defaultValue—Default value for a property. @minimum@maximum—Minimum and maximum value for an integer property. @minlength@maxlength—Length restriction on a string value property. @instanceId@array—A configuration object can have multiple instances and specify a property which uniquely identifies an instance. @group, @key, @component—Mechanism to identify the root level configuration objects which can read/written to config store layer **122**. @pattern—Mechanism to specify regex. @foreign_key—Mechanism to define co-relation between configuration objects. @unique—Mechanism to specify that a property value must be unique across all instances.

(25) The schema definition also specifies data types. The data types include: long (or int for integers), double (for floating point numbers), string, and Boolean. The schema definition of FIG. 3A is fed to a schema toolkit and the schema toolkit generates two files. The first is a configuration schema, one example of which is illustrated in FIG. 3B. The second is a default configuration schema (not shown).

(26) In the embodiments, config store layer **122** employs a validation engine to confirm that a value of a configuration property that is being written into key-value store **140** conforms to @minimum and @maximum specified for an integer type configuration property or @minlength and @maxlength for a string type configuration property. If a value falls outside these specified ranges, an error is returned in response to the API call that triggered the attempt to write the improper value into the key-value store **140**.

(27) FIG. 4A is another example of a schema definition. FIG. 4B is a configuration schema generated from the schema definition of FIG. 4A, and FIG. 4C is a default configuration schema generated from the schema definition of FIG. 4A. As shown in FIG. 4C, the default configuration property “layout” for a keyboard is “US Default.” The default configuration schema may be edited. FIG. 4D is an edited version of the default configuration schema of FIG. 4C. As shown in FIG. 4D, the default configuration property “layout” for a keyboard is now “Korean” (i.e., changed from “US Default” to “Korean”).

(28) FIG. 5E is an example that illustrates the use of an API to retrieve a default configuration. In response to this API, config store layer **122** retrieves the default configuration of the system service identified in the API from the default configuration schema, and returns the result. In the example illustrated in FIG. 5E, the default configuration is returned for “keyboard” and the property that has the default configuration is “layout.” As shown in FIG. 5E as the result, the default configuration property “layout” for a keyboard is “Korean.”

(29) FIG. 5F is an example that illustrates the use of an API to overwrite a default configuration. In the example of FIG. 5F, the set API is used to overwrite the default configuration, in particular to change the configuration property “layout” for a keyboard to “US Default.” In response to this API, config store layer **122** creates a configuration object for “keyboard” and updates key-value store **140** to add an entry to key-value store **140**. The added entry has “keyboard” as the key and “(layout:US Default)” as the value, indicating that the configuration property “layout” for a keyboard is “US Default.”

(30) FIG. 6 is a flow diagram that illustrates a system service that executes with a configuration that is set according to embodiments. The method of FIG. 6 begins when the system service is triggered at step **610**. Then, the system service at step **612** retrieves the default configuration of the system service from the default configuration schema and at step **614** retrieves the user configuration of the system service from key-value store **140**. At step **616**, the system service is executed with the default configuration and the user configuration. If there is any configuration property that is set in both the default configuration and the user configuration, the configuration property that is set in the user configuration overrides the configuration property that is set in the default configuration.

(31) FIG. 7 is a flow diagram that illustrates a method of resetting all system services to execute with a default configuration. The reset command may be issued, for example, as a reset API. The

method of FIG. 7 begins at step 710 at which config store layer 122 selects a configuration object. Then, at step 712, config store layer 122 deletes the configuration of the selected configuration object from key-value store 140. At step 714, if there are any more configuration objects, the process returns to step 710. If not, the process ends.

(32) FIG. 8 depicts entries in a key-value store 140 after resetting of FIG. 7 is carried out. As a result of the resetting, all configurations added to key-value store 140 are deleted. As a result, when the system services are run, the only configurations that will be applied are the default configurations defined in the default configuration schema. However, even after the resetting, any “vital” internal states for system services as designated in the schema definitions of the system services are retained in key-value store 140. Accordingly, the data “drift” is retained in key-value storage 140 because it is designated as a “vital” internal state in the schema definition for “ntp.”

(33) The embodiments described herein may employ various computer-implemented operations involving data stored in computer systems. For example, these operations may require physical manipulation of physical quantities. Usually, though not necessarily, these quantities may take the form of electrical or magnetic signals, where the quantities or representations of the quantities can be stored, transferred, combined, compared, or otherwise manipulated. Such manipulations are often referred to in terms such as producing, identifying, determining, or comparing. Any operations described herein that form part of one or more embodiments may be useful machine operations.

(34) One or more embodiments of the invention also relate to a device or an apparatus for performing these operations. The apparatus may be specially constructed for required purposes, or the apparatus may be a general-purpose computer selectively activated or configured by a computer program stored in the computer. Various general-purpose machines may be used with computer programs written in accordance with the teachings herein, or it may be more convenient to construct a more specialized apparatus to perform the required operations.

(35) The embodiments described herein may be practiced with other computer system configurations including hand-held devices, microprocessor systems, microprocessor-based or programmable consumer electronics, minicomputers, mainframe computers, etc.

(36) One or more embodiments of the present invention may be implemented as one or more computer programs or as one or more computer program modules embodied in computer readable media. The term computer readable medium refers to any data storage device that can store data which can thereafter be input to a computer system. Computer readable media may be based on any existing or subsequently developed technology that embodies computer programs in a manner that enables a computer to read the programs. Examples of computer readable media are hard drives, NAS systems, read-only memory (ROM), RAM, compact disks (CDs), digital versatile disks (DVDs), magnetic tapes, and other optical and non-optical data storage devices. A computer readable medium can also be distributed over a network-coupled computer system so that the computer readable code is stored and executed in a distributed fashion.

(37) Although one or more embodiments of the present invention have been described in some detail for clarity of understanding, certain changes may be made within the scope of the claims. Accordingly, the described embodiments are to be considered as illustrative and not restrictive, and the scope of the claims is not to be limited to details given herein but may be modified within the scope and equivalents of the claims. In the claims, elements and/or steps do not imply any particular order of operation unless explicitly stated in the claims.

(38) Virtualization systems in accordance with the various embodiments may be implemented as hosted embodiments, non-hosted embodiments, or as embodiments that blur distinctions between the two. Furthermore, various virtualization operations may be wholly or partially implemented in hardware. For example, a hardware implementation may employ a look-up table for modification of storage access requests to secure non-disk data.

(39) Many variations, additions, and improvements are possible, regardless of the degree of

virtualization. The virtualization software can therefore include components of a host, console, or guest OS that perform virtualization functions.

(40) Plural instances may be provided for components, operations, or structures described herein as a single instance. Boundaries between components, operations, and data stores are somewhat arbitrary, and particular operations are illustrated in the context of specific illustrative configurations. Other allocations of functionality are envisioned and may fall within the scope of the invention. In general, structures and functionalities presented as separate components in exemplary configurations may be implemented as a combined structure or component. Similarly, structures and functionalities presented as a single component may be implemented as separate components. These and other variations, additions, and improvements may fall within the scope of the appended claims.

Claims

1. A method of managing a configuration of a system service in each of a plurality of hosts, wherein each of the hosts is configured with a virtualization software for supporting execution of virtual machines therein, said method comprising: upon receiving an application program interface (API) call to operate on a configuration object for the system service that is backed by a configuration schema of the system service, updating a database in accordance with the configuration schema to update or store the configuration of the system service, wherein the system service executes with the configuration stored in the database, and wherein the configuration schema defines one or more configuration properties of the system service and one or more internal states of the system service, one of the internal states of the system service being designated by the configuration schema as an internal state of a first type that needs to be persisted in the database such that the internal state of the first type remains in the database upon reboot of the system service, and another one of the internal states of the system service being designated by the configuration schema as an internal state of a second type that does not need to be persisted in the database such that the internal state of the second type is lost upon reboot of the system service.
2. The method of claim 1, wherein the API call is a set API call to set the configuration of the system service, and upon receiving the API call, an entry for the configuration of the system service is added to the database.
3. The method of claim 1, wherein the API call is an update API call to update the configuration of the system service, and upon receiving the API call, an entry for the configuration of the system service in the database is updated.
4. The method of claim 1, further comprising: upon receiving a get API call to get the configuration of the system service, returning an entry for the configuration of the system service from the database.
5. The method of claim 1, further comprising: upon receiving a delete API call, deleting an entry for the configuration of the system service from the database.
6. The method of claim 1, wherein the configuration schema defines a range of permissible values for one of the configuration properties.
7. The method of claim 6, wherein an error is returned in response to the API call if a value for said one of the configuration properties, to be added to the database, is outside the range.
8. The method of claim 6, wherein an entry for the configuration of the system service in the database stores a value for the one of the internal states that is of the first type.
9. The method of claim 8, wherein the entry for the configuration of the system service in the database does not store a value of any internal state that is of the second type.
10. The method of claim 1, wherein the database is a key-value database, and each key corresponds to a unique configuration object.
11. A non-transitory computer readable medium comprising instructions to be executed in a

plurality of hosts that are each configured with a virtualization software for supporting execution of virtual machines therein, to carry out a method of managing a configuration of a system service in each of the plurality of hosts, said method comprising: upon receiving an application program interface (API) call to operate on a configuration object for the system service that is backed by a configuration schema of the system service, updating a database in accordance with the configuration schema to update or store the configuration of the system service, wherein the system service executes with the configuration stored in the database, and wherein the configuration schema defines one or more configuration properties of the system service and one or more internal states of the system service, one of the internal states of the system service being designated by the configuration schema as an internal state of a first type that needs to be persisted in the database such that the internal state of the first type remains in the database upon reboot of the system service, and another one of the internal states of the system service being designated by the configuration schema as an internal state of a second type that does not need to be persisted in the database such that the internal state of the second type is lost upon reboot of the system service.

12. The non-transitory computer readable medium of claim 11, wherein the API call is a set API call to set the configuration of the system service, and upon receiving the API call, an entry for the configuration of the system service is added to the database.

13. The non-transitory computer readable medium of claim 11, wherein the API call is an update API call to update the configuration of the system service, and upon receiving the API call, an entry for the configuration of the system service in the database is updated.

14. The non-transitory computer readable medium of claim 11, wherein the method further comprises: upon receiving a get API call to get the configuration of the system service, returning an entry for the configuration of the system service from the database.

15. The non-transitory computer readable medium of claim 11, wherein the method further comprises: upon receiving a delete API call, deleting an entry for the configuration of the system service from the database.

16. The non-transitory computer readable medium of claim 11, wherein the configuration schema defines a range of permissible values for one of the configuration properties.

17. The non-transitory computer readable medium of claim 16, wherein an error is returned in response to the API call if a value for said one of the configuration properties, to be added to the database, is outside the range.

18. The non-transitory computer readable medium of claim 11, wherein the database is a key-value database, and each key corresponds to a unique configuration object.

19. A cluster of hosts each configured with a virtualization software for supporting execution of virtual machines therein, wherein each of the hosts includes a processor and memory for executing a system service, and is configured to carry out a method of managing a configuration of the system service, said method comprising: upon receiving an application program interface (API) call to operate on a configuration object for the system service that is backed by a configuration schema of the system service, updating a database in accordance with the configuration schema to update or store the configuration of the system service, wherein the system service executes with the configuration stored in the database, and wherein the configuration schema defines one or more configuration properties of the system service and one or more internal states of the system service, one of the internal states of the system service being designated by the configuration schema as an internal state of a first type that needs to be persisted in the database such that the internal state of the first type remains in the database upon reboot of the system service, and another one of the internal states of the system service being designated by the configuration schema as an internal state of a second type that does not need to be persisted in the database such that the internal state of the second type is lost upon reboot of the system service.

20. The cluster of hosts of claim 19, wherein an entry in the database stores values of the one or

more configuration properties and the one of the internal states that is of the first type, and does not store a value of any internal state that is of the second type.
